

Assignment Name:

Understanding the Problem:

1. How did you approach understanding the challenge?

Clearly define the problem and the goal. Break down the problem into several steps/processes. If that's not enough then illustrate a visual or discuss it with peers.

2. Were there any parts of the problem that you found confusing at first? If so, how did you resolve the confusion?

None.

Planning the Solution:

1. Did you create a plan or break the problem into smaller steps before coding?

I broke the problem into three steps: Getting the input, calculating the outcome using the provided formula, and outputting the outcome.

2. How did you decide on which tools, algorithms, and data structures to use?

Functions: Used to organize the code and make the main function more readable, also provides reusability for getting inputs.

Conditionals: Used to check the input for the boolean, deciding whether to calculate the principal or return, and deciding what to print.

Try: Used for gathering inputs to prevent crashes from user error.

While Loops: Gives the user infinite attempts to input a proper value.

Implementation:

1. Did you write the code in smaller pieces or attempt the entire solution at once?

Smaller pieces, details mentioned above.

2. How did you test your solution to ensure that it was working correctly?

I ran the code using data given by the example in the textbook, if it ends up the same then it works properly.

Overcoming Challenges:

1. What part of the problem was the most difficult for you?

I first tried to make a general function for gathering inputs that accepted all types. Unfortunately, this wouldn't work that well with Java's static typing so I had to give up and divide the functions based on their type.

2. How did you handle moments where you felt stuck or unsure what to do next?

There weren't any for this project.

Learning:

1. Was there anything that you learned that you think will help you with future challenges?

I learned to not force dynamic typing implementations into Java and instead going with the flow and use static typing's strengths.